

Every Face Has a Story

Automated die studies at civilisational scale:
estimating the monetary output of the Roman Empire
from millions of unprovenanced coin images

1. Abstract

2. Introduction

3. Related Work

3.1. Die Studies in Numismatics

3.2. Deep Metric Learning for Cultural Heritage

4. Data

4.1. Training Data: Schaefer / RRD

Richard Schaefer’s Roman Republican Die Project (RRDP) provides our ground-truth labels: 77587 catalogued specimen-faces, spanning 16967 hand-identified dies across 445 RRC issues (7941 obverse, 9030 reverse). We have downloaded 70371 single-face crops (median 464×455 px; 10th-percentile short side 409 px) and trained the embedding on the bulk of this corpus.

The die-size distribution is the regime that governs output estimation: 34.5% of obverse dies are seen exactly once and 50.2% at most twice, while the largest die is represented by 72 coins. For benchmarking we use the 89 obverse issues with at least 50 coins and 15 dies (34175 coins). Near-duplicate imagery is handled by the dedup pass.

4.2. Deployment Data: sixbid archive

The deployment corpus is the sixbid auction archive (12.4M lots). A high-recall prefilter keeps 2.58M lots with any ancient/Roman signal; of these, 601528 lots cite a catalogue and were downloaded as full-resolution two-face photographs. Because no die study can proceed without knowing which coinage a coin belongs to, every lot is first assigned a **catalogue type**: a CRRO (RRC), OCRE (RIC) or RPC identifier. Citations in the lot description resolve 459958 lots to a type; those labels train an image model (shared DINOv2 ViT-B over obverse and reverse, cosine classifier, nearest-neighbour assignment against the labelled gallery) which reaches 0.94 / 0.823 / 0.646 held-out exact-type accuracy on RRC / RIC / RPC (top-5: 0.964 / 0.915 / 0.754), and refuses non-Roman coins (Greek, Byzantine, Celtic, world) at a false-annotation rate below 1%. Applied to the uncited remainder of the candidate pool it annotates a further 355949 lots, for 930759 typed coins in total: 175303 RRC (1543 types), 621147 RIC (20542 types) and 134309 RPC (13574 types). Figure 1 summarises the corpus. The type layer supplies the population each die study is run within, the dated denominator for every issue, and the structural match between the deployment pools and the Schaefer benchmark pools.

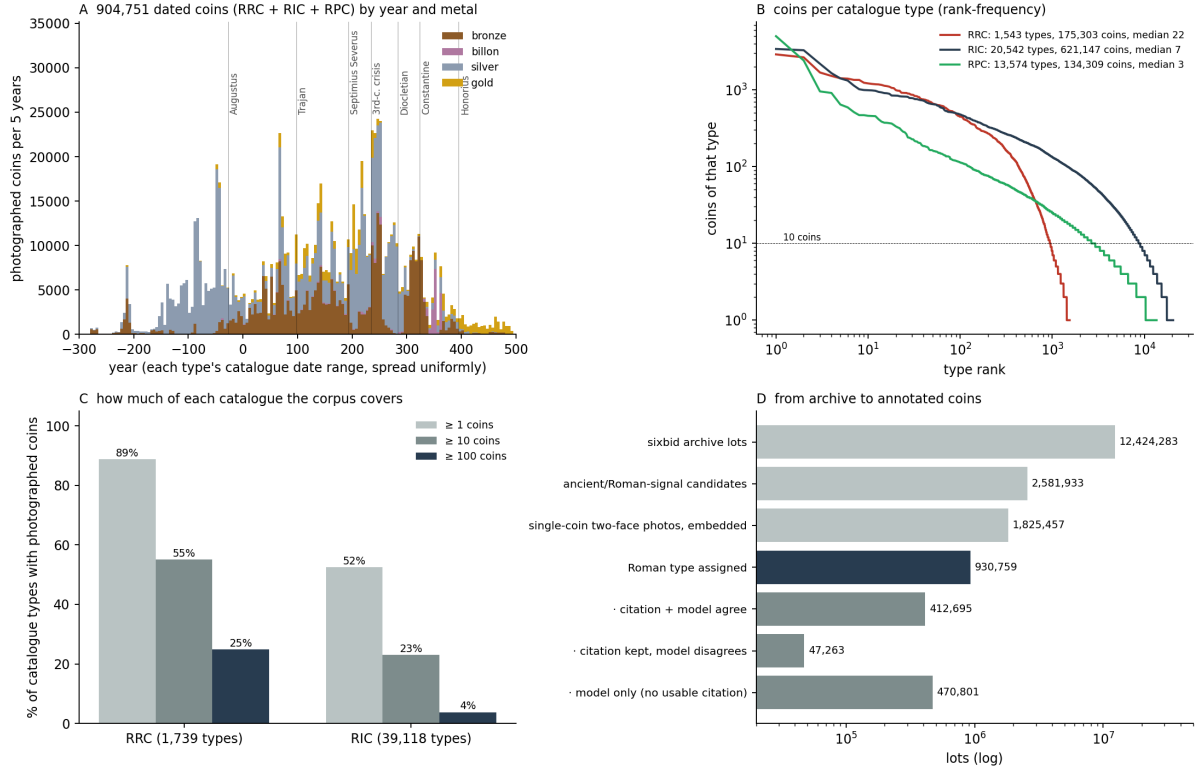


Figure 1: The annotated corpus. (A) Dated coins by 5-year bin and metal; each coin is spread uniformly over its type’s catalogue date range (interval data, no lot midpoints). RRC and RIC dates come from CRRO/OCRE; RPC Online publishes no bulk data, so its type records (province, city, reign, dating, metal) were collected page by page, with reign-level dates where a type has no explicit dating. (B) Coins per catalogue type: RRC is head-heavy (median 22 coins per type), RIC and RPC are long-tailed (medians 7 and 3). (C) Share of each catalogue’s types with at least 1, 10 and 100 photographed coins. (D) From archive lots to typed coins, by annotation source. “Coins” are auction lots; resales are not yet collapsed at this level.

5. Method

5.1. Segmentation

5.2. Embedding: DINOv3 + LoRA

5.3. Nearest-Neighbour Search

5.4. Die Clustering

5.5. Output Estimation

Given a die clustering, we estimate the number of dies originally used with coverage estimators that correct for never-sampled dies. Good–Turing coverage is $c = 1 - \frac{f_1}{n}$ (f_1 = dies seen once); Chao1 gives a lower bound $\hat{D} = d + \frac{f_1^2}{2f_2}$. As a sanity check on Schaefer’s own labels, the obverse corpus has coverage 0.9278 over 7941 observed dies, so Chao1 implies 10961 obverse dies were used in total. Crucially there is no absolute ground-truth die count: we validate the estimator by rarefaction, not against a known D . The output figure is $\hat{D} \times \text{coins-per-die}$, where the multiplier — not the algorithm — is the dominant uncertainty.

5.6. Counting Dies Without Clustering

Cluster counting fails in one specific way: a single false link merges two dies, and enough false links percolate into giant clusters, so single-linkage is only safe at an edge precision of 0.999. The quantity we

want can instead be read off pairs. Within a type whose dies carry shares p_i , two coins drawn at random share a die with probability $q = \sum p_i^2$, and $D_{\text{eff}} = \frac{1}{q}$ is the effective number of dies, the same object as the effective population size in genetics. Estimating q needs only the fraction of within-type pairs the verifier accepts, corrected by its recall and false-positive rate measured on Schaefer (leave-one-type-out); a false link is then a linear bias, never a merge. Figure 2 shows the calibration and the result: on the 10 bench types with at least 50 coins, the pairwise estimate sits at 0.917 of the truth with cv 0.238, flat across thresholds, against 1.648 (cv 0.217) for single-linkage at the same threshold. The residual spread is the verifier's per-type recall, which is the remaining lever.

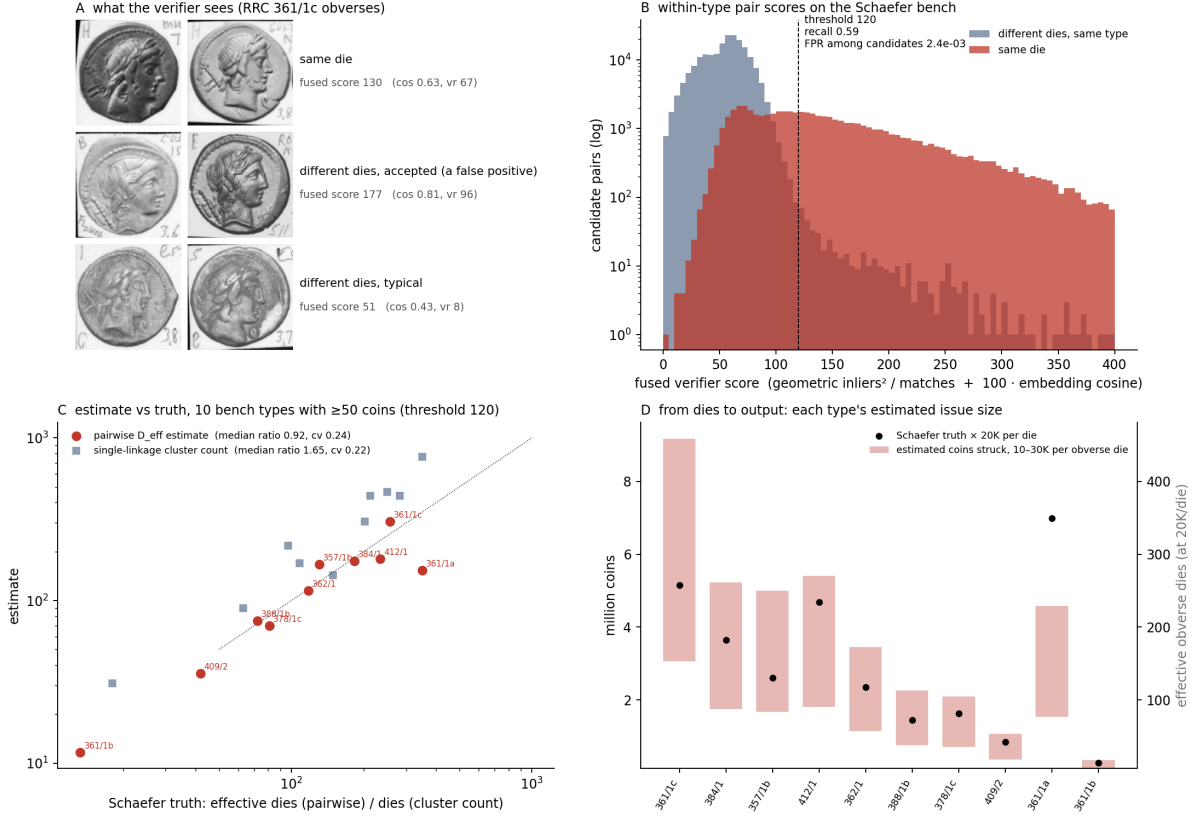


Figure 2: Counting dies without clustering. (A) Within one type, a same-die pair, a different-die pair the verifier accepts at the operating point, and a typical different-die pair, with fused scores. (B) Within-type pair scores on the Schaefer bench; at threshold 120 recall is 0.586 and 0.00244 of candidate pairs are false positives. (C) Pairwise D_{eff} estimate against Schaefer truth per type (leave-one-type-out calibration), with the single-linkage cluster count at the same threshold. (D) From dies to output: D_{eff} times the conventional 10000–30000 coins per obverse die, per type; the coins-per-die assumption dominates the uncertainty.

5.7. Republican Silver Output from Die Matches

Applying the pairwise estimator to the sixbid corpus gives the first die-based output series for the whole Republican denarius coinage, with every source of uncertainty propagated. Every same-type candidate pair among the 99050 RRC-typed lots was geometrically verified (1441292 pairs, both faces). Three properties of an auction corpus had to be measured first. Coins are resold: lot pairs with identical published weights (to 0.01 g, against a 7% chance rate) show that pairs scoring above 800 are the same specimen; collapsing them leaves 90265 specimens, and the residual resale fraction per score band, read off the same weight mixture, is subtracted. Recall depends on photograph quality: same-die pairs of modern photographs score about twice what Schaefer's catalogue scans do (BnF–BnF pairs are recovered at 0.98, scan–scan at 0.59), so the scan-domain recall of 0.47 cannot be used; the bench's photograph–photograph pairs give 0.689, and a corpus-internal measurement (a resold coin's second photograph against its die-mate) gives 0.97, and recall is treated as uniform between them. And the sample is not die-clustered:

same-house pairs share dies only marginally more than chance, and the pair-share rate from cross-house pairs alone agrees with the full estimate to 1%.

The model is the one population genetics uses for effective population size. A type's D dies each strike $c_i \sim \text{Gamma}(\alpha, \frac{\mu}{\alpha})$ coins; two coins share a die with probability $q = \sum p_i^2 \approx \frac{1+\frac{1}{\alpha}}{D}$, so the pair-share rate measures the *effective* die count $D_{\text{eff}} = \frac{D}{1+\frac{1}{\alpha}}$ and output is $T = (1 + \frac{1}{\alpha}) \frac{\mu}{q}$. Schaefer's per-die counts fit a negative binomial with $\alpha = 2.68$ (Figure 3 C), so $D = 1.37 D_{\text{eff}}$. The accepted-pair count is binomial in q , the recall and the false-positive rate; coins per obverse die is a log-normal prior spanning 10000–30000 (the numismatic convention); each type's weight is bootstrapped from the upper quartile of its own lots' published weights, fineness is 0.96 ± 0.015 , and each type's production year is uniform over its catalogue date range. On the 4 Schaefer types present in the corpus the estimate runs at 0.695 of Schaefer's count; the disagreement is not divided out but carried as a symmetric systematic (log-sd 0.363), so it widens the interval rather than moving the estimate. All are sampled jointly in 4000 draws.

Figure 3 gives the posterior for 499 silver denarius types with at least 20 specimens: 2660 tonnes of silver struck between 211 and 31 BC (95%: 1058–6812 50%: 1929–3660), peaking at 478 tonnes per decade in the –90 s. The two surges, the Social War and the civil wars of the 40s, are the ones the literature expects. The variance decomposition (Figure 3 B) is the point: 55.9% of the posterior log-variance is the unexplained disagreement with Schaefer and 35.3% the coins-per-die prior, while pair sampling, verifier calibration, weights and dating together contribute almost nothing. The die count is measured; what remains is one open discrepancy and one numismatic convention, and the figure says which to work on.

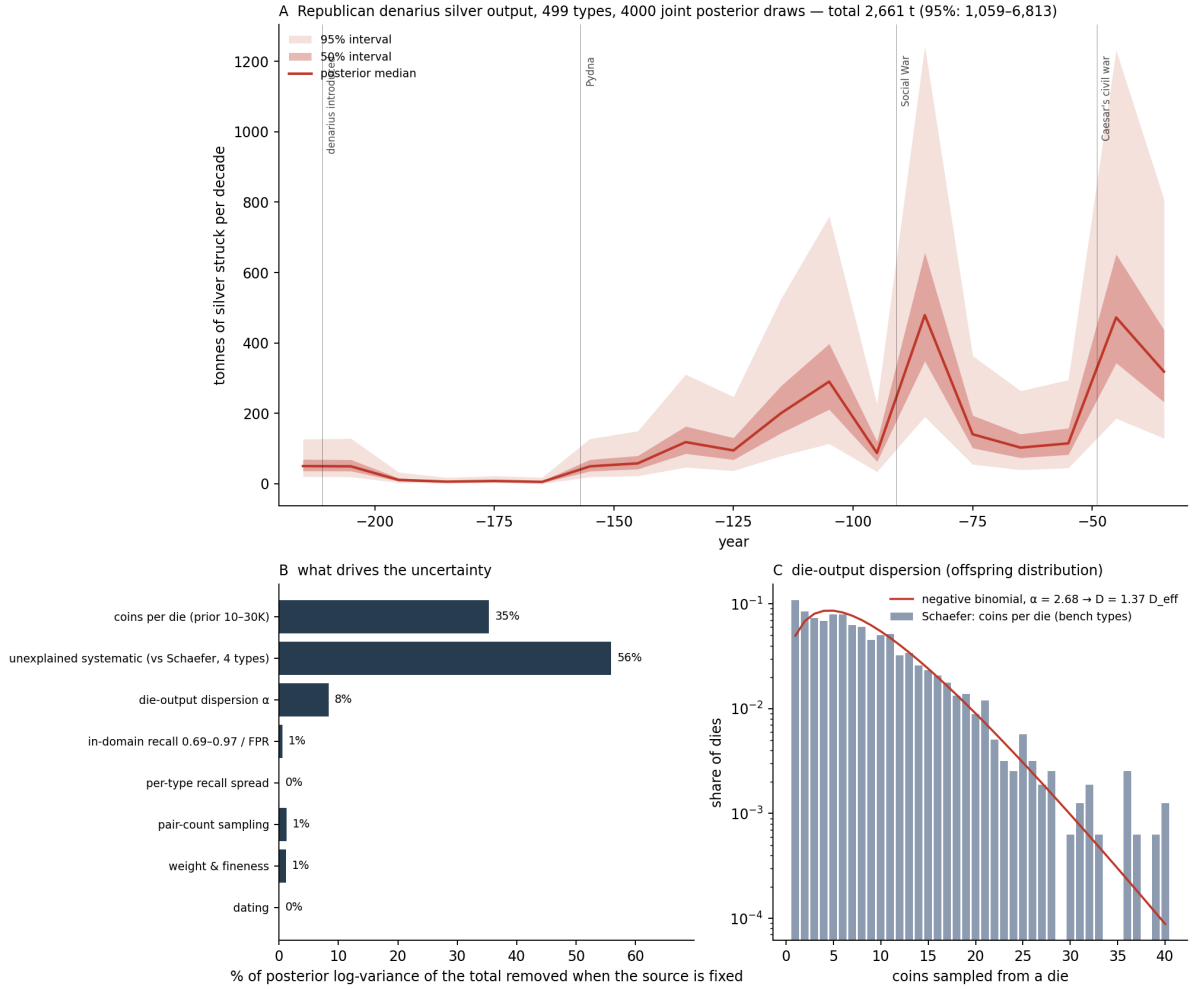


Figure 3: Republican silver output from die matches. (A) Tonnes of silver struck per decade: posterior median with 50% and 95% intervals from 4000 joint draws over every source of uncertainty. (B) Share of the posterior log-variance of the total removed when each source is fixed at its centre. (C) Schaefer’s coins-per-die distribution on the bench types and the negative-binomial fit giving the die-output dispersion α , hence $D = 1.37D_{\text{eff}}$.

6. Experiments

6.1. Validation Levels

7. Results

On the frozen RRDP test split, geometric verification fused with the DINOv3-LoRA embedding separates die-mates from hard (same-type) negatives with verification AUC 0.9591 (obverse) and 0.9476 (reverse). Pairwise quality is high; converting trustworthy links into the correct die *count* — robust to the percolation cliff — remains the binding constraint (see Discussion).

8. Discussion

9. Conclusion

Bibliography